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Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients

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1. INTRODUCTION

Applied physics in medicine, particularly Magnetic Resonance Imaging (MRI), has revolutionized diagnostics by enabling non-invasive detection of structural abnormalities. Epilepsy is a brain disorder characterized by a predisposition to generate epileptic seizures [1], with an estimated prevalence of 180–400 affected Mexican children per 100,000 inhabitants.

Current literature reports corpus callosum (CC) atrophy in epilepsy patients [2], which compromises interhemispheric connectivity. However, few studies have characterized CC volumetrics in Mexican pediatric populations using MRI.

2. OBJETIVE

This study quantitatively assesses corpus callosum volume loss in pediatric epilepsy through MRI volumetry with statistical analysis.

3. MATERIALS AND METHODS

This case-control study included 30 pediatric patients with epilepsy and 30 healthy controls, matched 1:1 by sex and age (range: 4-15 years).

T1-weighted MPRAGE sequences were acquired for all participants on a Siemens Skyra 3T scanner. Key acquisition parameters included a repetition time (TR) of 2300 ms, echo time (TE) of 2 ms, inversion time (TI) of 930 ms, a flip angle of 10°, and an isotropic voxel size of 1.0 x 1.0 x 1.0 mm³.

Post-processing and volumetric measurements of the corpus callosum subregions were performed using FreeSurfer software [3]. The resulting normalized volumetric data were subjected to statistical analysis in R [4] to identify significant correlations and differences between groups, aiming to detect specific patterns of regional atrophy that distinguish patients from controls.

4. RESULTS

Total intracranial volume (TIV) was compared between healthy controls and epilepsy patients to account for overall brain size differences in subsequent analyses.

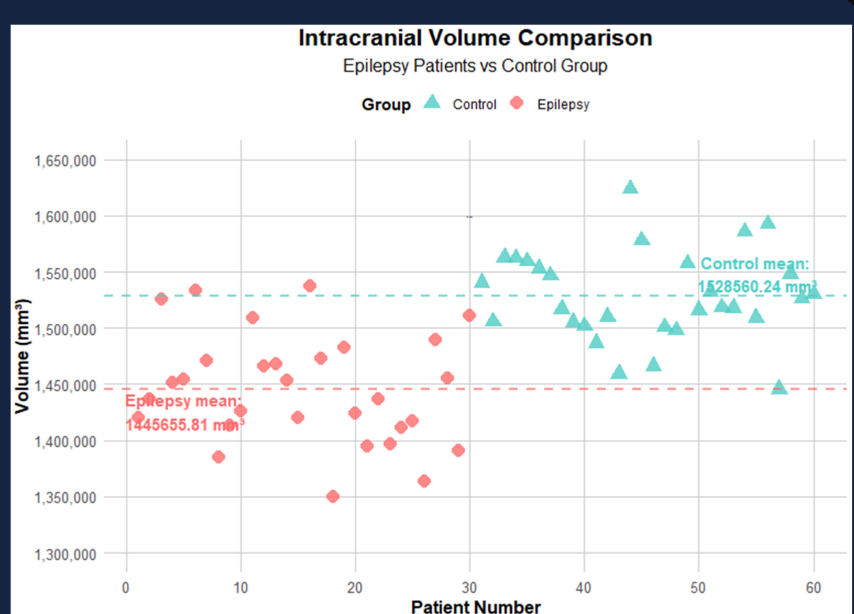

Figure 1. Intracranial volume comparison

Utilizing segmentation data derived from FreeSurfer and 3D Slicer® for enhanced three-dimensional visualization, a comparative 3D reconstruction of the corpus callosum was generated.

Figure 2 presents a simultaneous visualization of the segmented corpus callosum, depicted in blue, with the healthy control shown on the left and the representative epilepsy patient on the right.

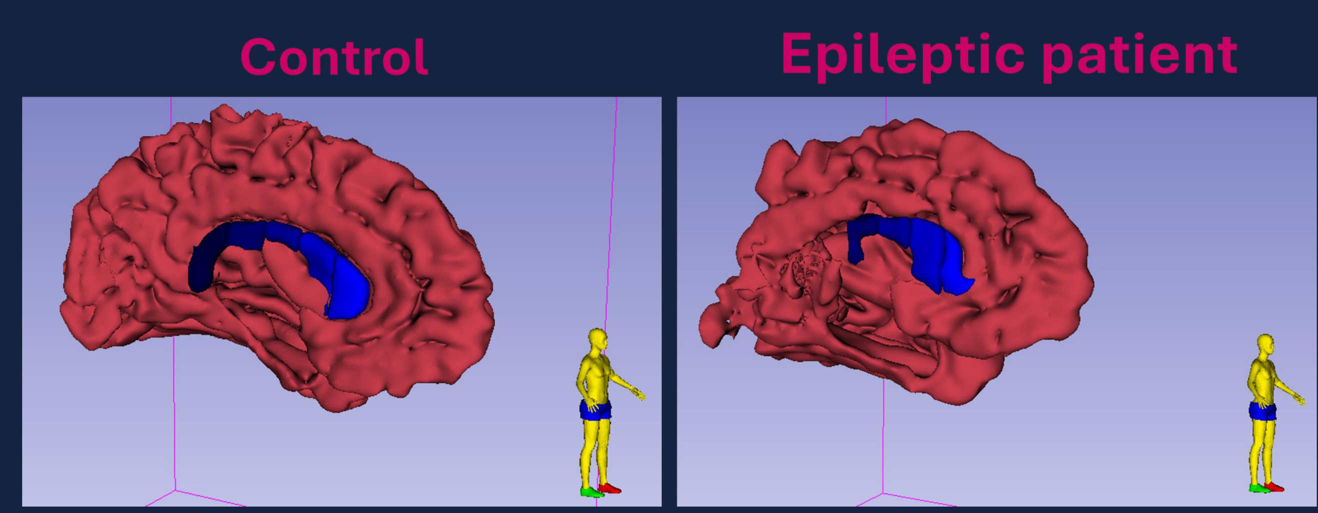

Figure 2. Comparative 3D Reconstruction of the Corpus Callosum.

Statistical evaluation, performed using Student's t-test in the R environment [4], identified a statistically significant reduction in volume specifically within the posterior segments of the corpus callosum in the pediatric epilepsy cohort ($p=0.03$).

This region-specific finding suggests a targeted vulnerability to epileptic activity or its sequelae, moving beyond generalized measures to reveal a structurally meaningful and localized difference between patients and matched controls. This result provides quantitative evidence supporting the observed qualitative morphological alterations.

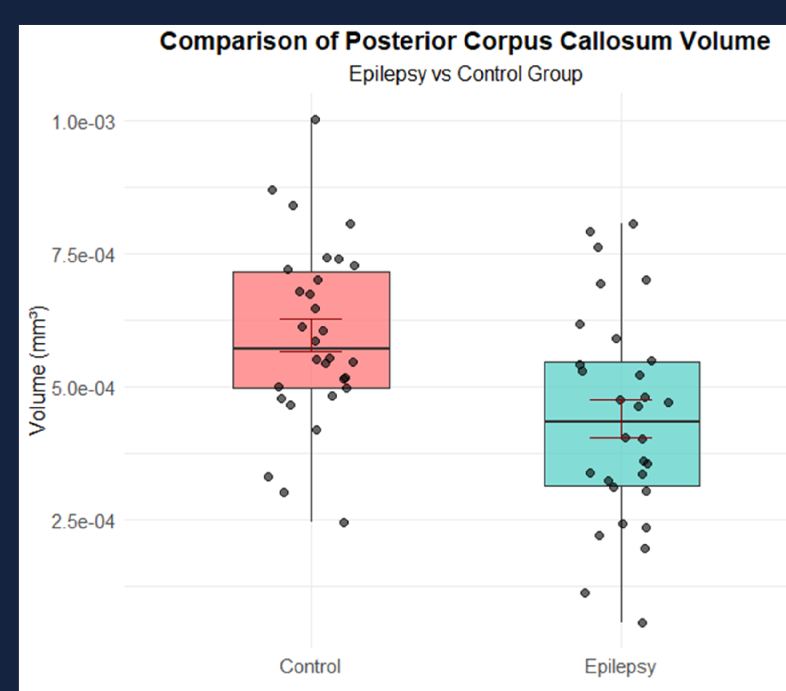

Figure 3. Comparisom of Posterior Corpus Callosum Volume

5. CONCLUSIONS

Our findings demonstrate that posterior corpus callosum (CC) volume reduction in epilepsy patients may impair interhemispheric communication and correlate with treatment-resistant seizures. While these results are statistically significant ($p=0.03$), their clinical relevance requires validation through larger multicenter studies. This work particularly highlights the critical need for dedicated neuroimaging research in pediatric epilepsy within local populations.

7. REFERENCES

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